

Introduction

This paper looks at a classic question in labor economics: does raising the minimum wage actually reduce employment? Basic supply and demand theory says yes. When the price of labor goes up employers hire fewer workers. But that prediction has been challenged repeatedly and the most famous challenge came from Card and Krueger in 1994. Their study of New Jersey's 1992 minimum wage increase found employment at fast food restaurants actually went up after the raise. This paper replicates and extends that analysis using the original Card and Krueger dataset. The goal is to estimate the causal effect of New Jersey's minimum wage increase on full time equivalent employment at fast food restaurants. This matters because minimum wage policy is one of the most debated tools in labor economics and the empirical evidence still shapes real decisions about worker pay.

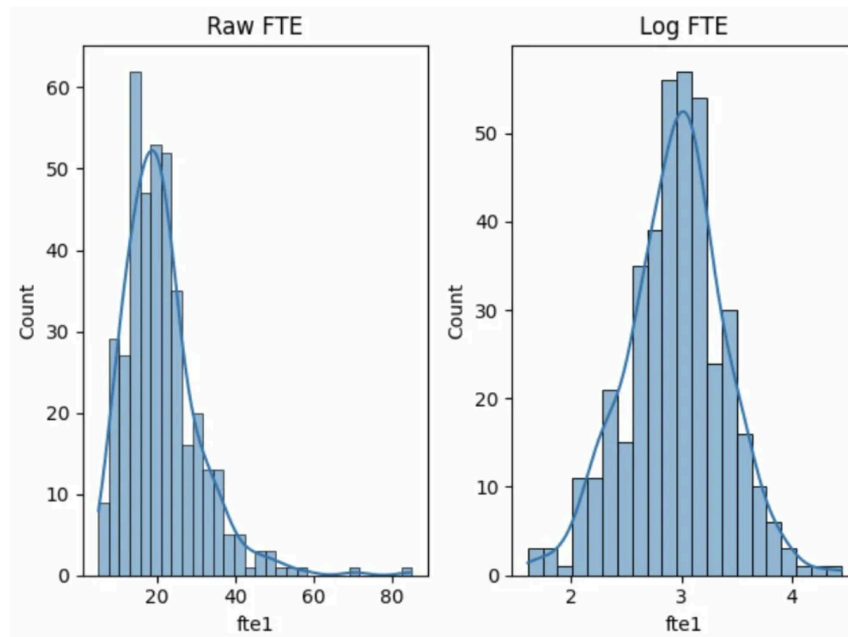
Data Description

The data comes from Card and Krueger's original two wave survey of fast food restaurants. The first wave was collected in early 1992 before New Jersey raised its minimum wage from \$4.25 to \$5.05. The second wave came after the increase took effect later that year. The survey covered restaurants in New Jersey (the treated state) and eastern Pennsylvania (the control). Pennsylvania did not change its minimum wage during this period which is what makes it a valid comparison group.

The chains included are Burger King, KFC, Roy Rogers, and Wendy's. After filtering for restaurants still open in the second wave and dropping rows with missing values in the key variables the working sample is 359 observations as confirmed in both regression outputs. The outcome variable is full time equivalent employment, calculated as full time employees plus managers plus half of part time employees. This was constructed for both waves and the

change between them is `delta_fte`. The main treatment indicator NJ equals 1 for New Jersey and 0 for Pennsylvania.

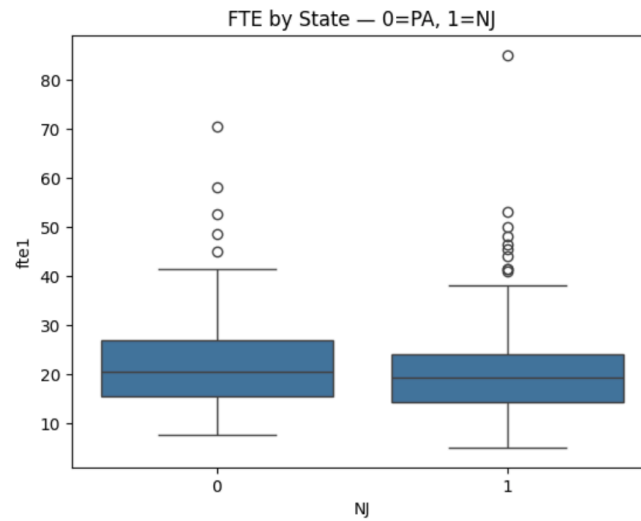
Raw FTE vs Log FTE:



Looking at raw pre-period FTE levels shows the distribution is right skewed. Most restaurants cluster between 15 and 25 employees and the tail stretches out past 80. The log transformation brings the distribution much closer to symmetric, centering it around 3 on the log scale.

The boxplot below compares pre-period FTE levels across the two states. Pennsylvania restaurants (NJ = 0) have a median around 20, an interquartile range roughly from 15 to 27, and whiskers reaching up to about 42. New Jersey restaurants (NJ = 1) look very similar with a median just under 20, an IQR roughly from 15 to 24, and whiskers reaching up to about 39. Both groups have a handful of high-end outliers. Pennsylvania has several above 45 including one near 70 and New Jersey has a few above 45 including one near 85. The two distributions overlap substantially which supports the comparability of the groups before treatment.

FTE by State (0=PA, 1=NJ):



Outlier analysis using the IQR method finds a lower fence of -0.34375 and an upper fence of 39.40625 applied to delta_fte. There are 17 restaurants that fall outside these bounds. These are restaurants with unusually large employment swings between the two survey waves and their presence in the sample could inflate variance.

Table 1 below summarizes the key variables by treatment group comparing NJ and PA restaurants at baseline.

Table 1. Summary Statistics by Treatment Group

	fte1		wage_st		delta_fte	
	mean	std	mean	std	mean	std
NJ						
0	23.39	11.92	4.63	0.35	-2.06	10.76
1	20.60	9.14	4.61	0.34	0.66	8.39

Methodology

The identification strategy is Difference in Differences. The core idea is that any general trend in fast food employment would show up in both states. Comparing the change in employment from before to after in NJ against the same change in PA isolates what the minimum wage hike did beyond that common trend. The DiD estimator is captured by the NJ coefficient in an OLS regression where the outcome is `delta_fte`.

The main equation is $\text{delta_fte} = b_0 + b_1(\text{NJ}) + b_2(\text{chain}) + b_3(\text{co_owned}) + b_4(\text{wage_st}) + e$

NJ is the treatment indicator and b_1 is the estimate of interest. Chain fixed effects are included because different fast food chains may have systematically different staffing norms. The `co_owned` variable controls for whether a location is company operated versus franchised. Starting wage is included to capture how exposed each restaurant was to the new minimum since restaurants already paying well above \$4.25 had less reason to adjust employment.

The key assumption for DiD is parallel trends. Without the wage increase NJ and PA employment would have followed the same trajectory. The boxplot above provides informal support for this. The pre-period employment distributions in both states are nearly identical in median and spread which makes it reasonable to treat Pennsylvania as a valid counterfactual for New Jersey.

All models use HC1 heteroscedasticity robust standard errors. Employment data at the restaurant level is inherently noisy so robust inference is necessary.

Results

OLS Regression Results (Model 1):

OLS Regression Results						
Dep. Variable:	delta_fte	R-squared:	0.030			
Model:	OLS	Adj. R-squared:	0.014			
Method:	Least Squares	F-statistic:	1.940			
Date:	Sun, 22 Mar 2026	Prob (F-statistic):	0.0736			
Time:	21:24:16	Log-Likelihood:	-1278.8			
No. Observations:	359	AIC:	2572.			
Df Residuals:	352	BIC:	2599.			
Df Model:	6					
Covariance Type:	HC1					
	coef	std err	z	P> z	[0.025	0.975]
Intercept	9.1081	6.213	1.466	0.143	-3.069	21.285
C(chain) [T.2]	0.3927	1.085	0.362	0.717	-1.734	2.520
C(chain) [T.3]	-1.8167	1.168	-1.555	0.120	-4.106	0.473
C(chain) [T.4]	-0.0883	1.639	-0.054	0.957	-3.300	3.124
NJ	2.2845	1.372	1.666	0.096	-0.404	4.973
co_owned	0.5393	0.898	0.601	0.548	-1.220	2.298
wage_st	-2.3408	1.417	-1.652	0.099	-5.118	0.436
Omnibus:	39.700	Durbin-Watson:	2.184			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	106.616			
Skew:	-0.512	Prob(JB):	7.06e-24			
Kurtosis:	5.466	Cond. No.	68.2			

The baseline model estimates the NJ coefficient at 2.2845 with a standard error of 1.372. New Jersey restaurants saw roughly 2.28 more full time equivalent employees on average relative to Pennsylvania after the wage increase. The p-value is 0.096 and the 95 percent confidence interval runs from -0.404 to 4.973. The result is significant at the 10 percent level. Employment did not fall in the treated state.

The starting wage variable has a coefficient of -2.3408 with a standard error of 1.417 and a p-value of 0.099. Restaurants that were already paying higher wages saw larger employment drops on average. Those restaurants likely faced more direct pressure from the new floor. The chain dummies are not significant. Chain 3 shows the largest magnitude at -1.8167 with a

p-value of 0.120 but does not clear conventional thresholds. The R-squared is 0.030 and adjusted R-squared is 0.014.

OLS Regression Results (Model 2):

OLS Regression Results						
Dep. Variable:	delta_fte	R-squared:	0.030			
Model:	OLS	Adj. R-squared:	0.011			
Method:	Least Squares	F-statistic:	1.768			
Date:	Sun, 22 Mar 2026	Prob (F-statistic):	0.0927			
Time:	21:24:16	Log-Likelihood:	-1278.8			
No. Observations:	359	AIC:	2574.			
Df Residuals:	351	BIC:	2605.			
Df Model:	7					
Covariance Type:	HC1					
	coef	std err	z	P> z	[0.025	0.975]
Intercept	9.1604	6.240	1.468	0.142	-3.070	21.391
C(chain) [T.2]	0.3555	1.092	0.326	0.745	-1.784	2.496
C(chain) [T.3]	-1.8421	1.173	-1.571	0.116	-4.141	0.457
C(chain) [T.4]	-0.1361	1.606	-0.085	0.932	-3.285	3.012
NJ	2.4944	1.968	1.267	0.205	-1.363	6.351
co_owned	1.0317	2.450	0.421	0.674	-3.769	5.833
NJ:co_owned	-0.5964	2.557	-0.233	0.816	-5.608	4.415
wage_st	-2.3853	1.466	-1.628	0.104	-5.258	0.487
Omnibus:	39.148	Durbin-Watson:	2.186			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	105.242			
Skew:	-0.503	Prob(JB):	1.40e-23			
Kurtosis:	5.454	Cond. No.	68.4			
Notes:						
[1] Standard Errors are heteroscedasticity robust (HC1)						

The second model adds an interaction between NJ and co_owned. The interaction coefficient is -0.5964 with a standard error of 2.557 and a p-value of 0.816. The main NJ coefficient rises slightly to 2.4944 with a standard error of 1.968 and a p-value of 0.205. The R-squared stays at 0.030 but adjusted R-squared drops to 0.011. Adding the interaction absorbs a degree of freedom without improving fit. There is no evidence that franchise versus company-owned restaurants responded differently to the wage increase. Table 2 below places both models side by side.

Table 2 (Model 1 vs Model 2 side by side):

	Model 1	Model 2
Intercept	9.1081 (6.2131)	9.1604 (6.2401)
C(chain) [T.2]	0.3927 (1.0853)	0.3555 (1.0919)
C(chain) [T.3]	-1.8167 (1.1682)	-1.8421 (1.1729)
C(chain) [T.4]	-0.0883 (1.6389)	-0.1361 (1.6065)
NJ	2.2845* (1.3715)	2.4944 (1.9679)
co_owned	0.5393 (0.8975)	1.0317 (2.4495)
wage_st	-2.3408* (1.4169)	-2.3853 (1.4655)
NJ:co_owned		-0.5964 (2.5569)
R-squared	0.0302	0.0303
R-squared Adj.	0.0136	0.0110

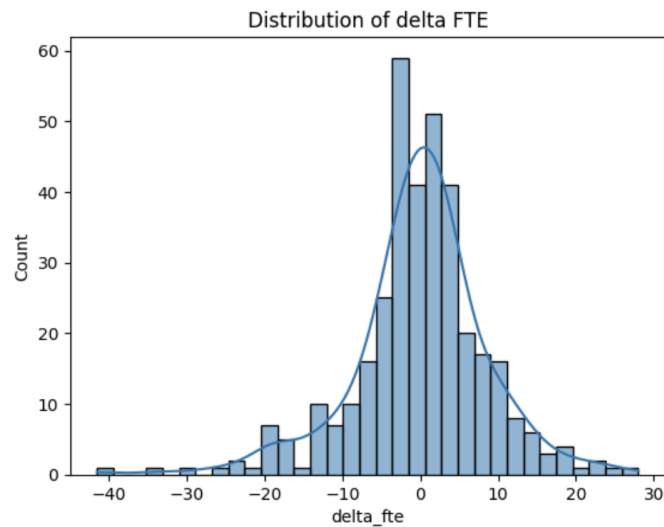
Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

The NJ estimate barely moves between specifications, from 2.28 to 2.49, which is a sign of stability rather than fragility.

Model Diagnostics

The distribution of delta_fte is roughly bell shaped but not symmetric. The histogram shows a longer left tail reaching as far as negative 40 and a shorter right tail topping out near positive 30. Most observations cluster between negative 10 and positive 10. The Jarque-Bera statistic from Model 1 is 106.616 with a p-value of 7.06e-24 which formally rejects normality. Skew is -0.512 and kurtosis is 5.466. Model 2 is essentially identical with a Jarque-Bera of 105.242, skew of -0.503, and kurtosis of 5.454.

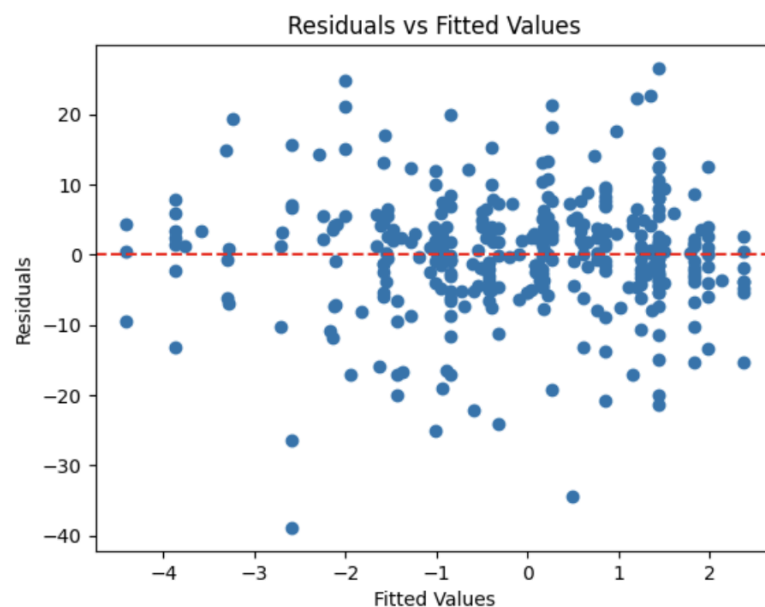
Distribution of delta FTE:



The result is mostly normal and non-normality is not a concern. OLS remains consistent and the HC1 robust standard errors handle the variance issue so inference is still valid. A log transformation was considered but delta_fte contains negative values so it cannot be directly applied. The change specification is also just the natural DiD approach regardless.

The Durbin-Watson statistic is 2.184 in Model 1 and 2.186 in Model 2. Both sit very close to 2 which means no meaningful serial autocorrelation in the residuals.

Residuals vs Fitted Values:



The residuals vs fitted values plot shows points scattered randomly around the zero line with no clear curve or pattern. This means there is no obvious non-linear relationship the model is missing. The spread of residuals is roughly consistent across fitted values though a few extreme negative residuals near -40 stand out on the left side. Those correspond to the same outlier restaurants flagged in the IQR analysis. Overall the plot does not reveal any structural problem with the model.

The raw FTE distribution shown earlier confirms the right skew in levels. Most restaurants have between 15 and 25 employees and a small number of high employment locations stretch the distribution toward 85. The log transformation shown alongside it produces a much more symmetric shape centered around 3, which suggests a log transformation of levels would have been appropriate had the research design called for a levels regression rather than a first-difference DiD.

The outlier analysis flags 17 restaurants outside the IQR fence (lower fence -0.34375, upper fence 39.40625). These are locations with unusually large changes in FTE between waves and could be inflating variance in the residuals.

Robustness Checks

The NJ coefficient is stable across both specifications, moving from 2.2845 in Model 1 to 2.4944 in Model 2 without changing sign or losing economic significance. The wage_{st} coefficient is equally stable going from -2.3408 to -2.3853 across the two models. Estimates that do not move much when the specification changes are generally more credible.

The 17 outlier restaurants identified by the IQR analysis are another robustness concern. These observations have delta_fte values beyond the fence of 39.41 or below -0.34 and represent unusually large employment swings. Dropping them from the sample would tighten the residual

distribution and likely reduce standard errors without changing the direction of the NJ coefficient since the effect appears across the bulk of the sample rather than being driven by a few extreme observations.

The interaction term result in Model 2 (NJ:co_owned = -0.5964, $p = 0.816$) also rules out the possibility that the positive NJ effect is concentrated in one ownership type. The effect is general. And the adjusted R-squared declining from 0.014 to 0.011 when the interaction is added confirms the baseline model is the right specification.

Conclusion

This analysis finds that New Jersey's minimum wage increase did not reduce fast food employment. The DiD estimate of roughly 2.28 additional full time equivalent employees in New Jersey relative to Pennsylvania is positive and marginally significant in the baseline model. This replicates the core direction of Card and Krueger's original finding. The null hypothesis that raising the minimum wage reduces employment is not supported by this data.

There are real limitations. The dataset covers only fast food restaurants in one corridor along the NJ and PA border. It cannot speak to larger wage increases or different labor markets. The parallel trends assumption is plausible given the near-identical pre-period distributions visible in the boxplot but it is still untestable with only two waves of data. The models also explain only about 3 percent of the variation in employment changes which means a lot of what drives individual restaurant decisions is unaccounted for.

Still the finding is consistent with the broader literature suggesting moderate minimum wage increases have small or even positive employment effects when employers have some degree of market power over workers. This replication confirms that conclusion holds with standard controls and robust inference applied.

Code Appendix

All analysis was conducted in Python using pandas, numpy, statsmodels, and seaborn. The code was written manually in Google Colab.